# A3-PS002-TC427

## Purpose-Based Consent Recording & Withdrawal Prototype

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# Problem Description

**Consent is often captured — but not understood.**

The gap is not the checkbox. It is the missing link between a decision, a purpose, and enforceable system behavior.

- Consent records may not clearly link decisions to specific purposes.
- Users may not understand which processing activities they agreed to.
- Withdrawal controls can be difficult to find or execute.
- Grant, update, and withdrawal events may not be tracked consistently.
- Processing services may rely on stale consent status.
- Auditing consent history is difficult without reliable records.

**How can we record purpose-specific consent and enable withdrawal with reliable, auditable history?**

# Background

**Purpose-based consent connects user decisions to specific processing activities and enforceable system behavior.**

A consent record is more useful when it identifies the purpose, the applicable notice or policy version, the time of the decision, and the current status. The prototype therefore treats consent as a lifecycle rather than a one-time checkbox.

The design separates the current consent state from the historical event log. This supports status checks during processing while preserving traceability for grants, changes, and withdrawals.

The next pages describe related work, the proposed solution, and the design details. [1]

# Literature & RFC Review

**Relevant legal principles and engineering guidance inform the prototype, without treating recommendations as proof of compliance.**

Digital Personal Data Protection Act, 2023 (India) provides relevant context for notice, consent, and user control where applicable.

GDPR Articles 5, 7, and 17 are useful reference points for purpose limitation, conditions for consent, and erasure rights where relevant.

OWASP guidance informs API security, access control, secure application design, and protection of state-changing operations. [2]

# Proposed Solution

**A purpose-based consent management prototype makes each decision understandable, traceable, and enforceable.**

The system provides a user-facing consent settings page, a backend API, a centralized consent management service, and an authoritative record of current status and historical events.

- Separate Allow / Decline choices for Essential Service Notifications, Product Updates, and Analytics.
- Clear purpose descriptions connected to the relevant processing activity.
- Timestamp records with purpose ID, policy version, and user reference.
- A dashboard for reviewing choices and withdrawing consent for one purpose.
- A status check before processing that depends on consent.
- Explicit confirmation only after the withdrawal event is persisted.

Example: a user may allow essential notifications, decline promotions, and later withdraw analytics consent. The prototype updates the authoritative analytics state and blocks future analytics processing that depends on that consent.

# Solution Architecture & Design

**Visual: architecture and enforcement boundaries for purpose-based consent.**

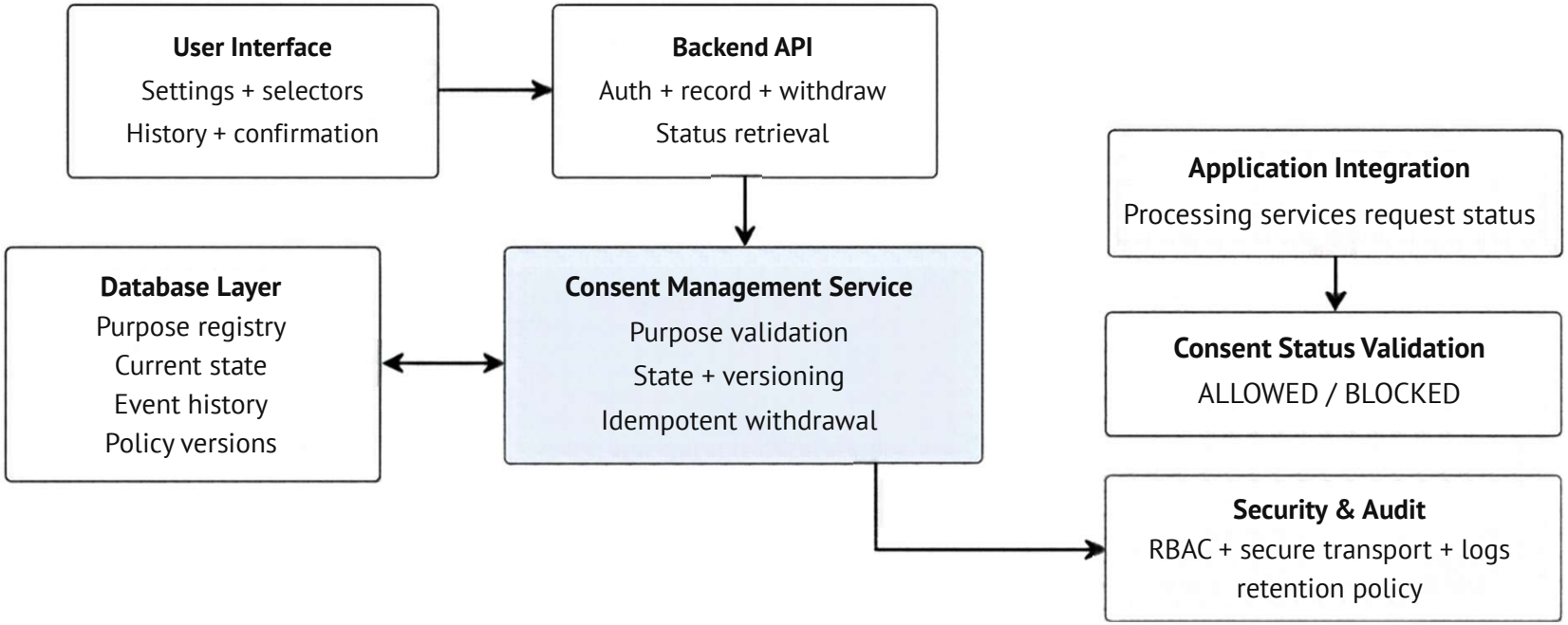
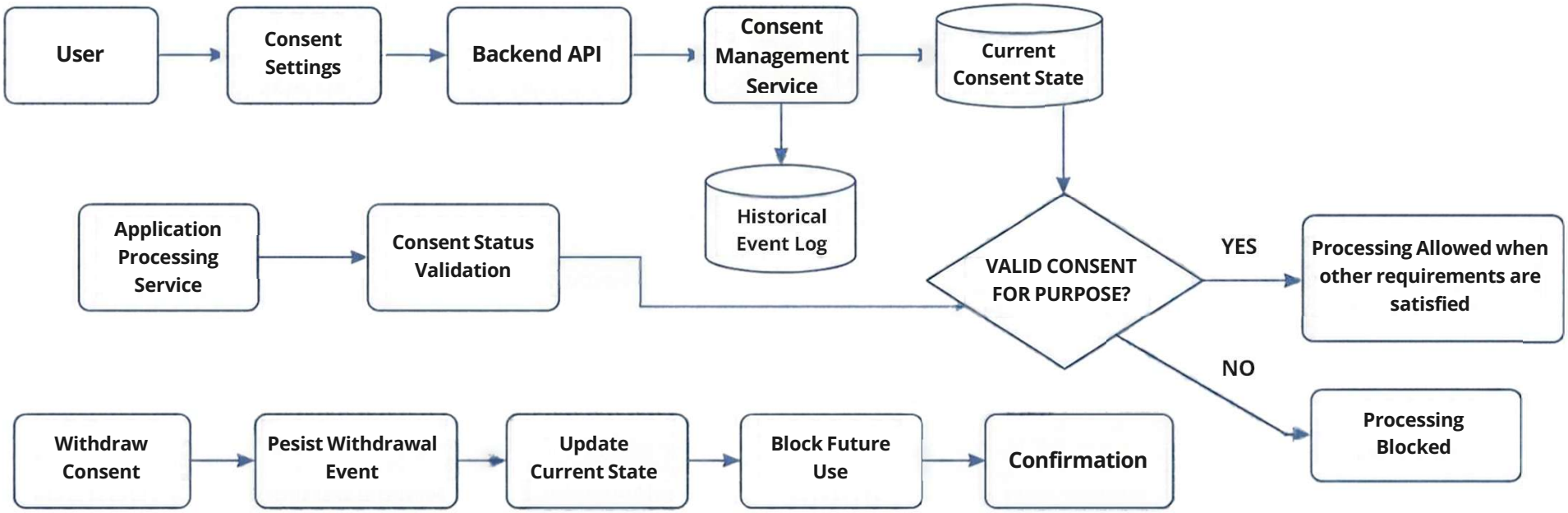



**Figure:** Purpose-Based Consent Architecture

Withdrawal updates the authoritative state first; integrated processing systems fetch the new status reliably.

# Data Flow Diagram

**Overview of the consent decision and enforcement flow.**




Failed writes and unavailable services preserve the prior state and return an explicit retry outcome.

**Figure:** Purpose-Based Consent Data Flow Diagram (DFD)

# Deployment Timeline

**A four-week implementation plan with security and usability validation built into the workstream.**

01

### WEEK 1 / Foundation

Requirements analysis; Purpose definitions; User journey mapping; Data model design.

02

### WEEK 2 / Core build

Consent interface; Backend API; Consent recording.

03

### WEEK 3 / Control & test

Withdrawal workflow; Status validation; Audit logging; Security testing.

04

### WEEK 4 / Integrate & deliver

Integration and usability testing; Bug fixes; Demonstration preparation; Final documentation.

# Bibliography I

[1] Digital Personal Data Protection Act, 2023 (India). Official Ministry of Electronics and Information Technology resources. https://www.meity.gov.in/data-protection-framework

[2] Regulation (EU) 2016/679 (General Data Protection Regulation), Articles 5, 7, and 17. https://eur-lex.europa.eu/eli/reg/2016/679/oj

[3] OWASP Foundation. API Security and secure application design resources. https://owasp.org

[4] Technology stack documentation: [Insert official programming language, framework, database, and testing-tool URLs after the implementation stack is selected].

Legal requirements and engineering recommendations are presented separately.

**Consent should be specific, traceable, and easy to withdraw.**

Purpose-specific choices • Reliable recording • Purpose-level withdrawal
Current-state enforcement • Traceable lifecycle events

## NEXT STEPS

**01** Complete the core prototype

**02** Test consent and withdrawal workflows

**03** Validate enforcement in integrated services

**04** Review security, usability, and privacy safeguards

**05** Prepare the live demonstration and collect feedback

# Thank You

Questions?